Predictive Modeling for Imbalanced Customer Transactions: An Optimized LightGBM Pipeline

EC452 Machine Learning Semester Project (Spring 2026)

Muhammad Taha (417609)
Department of Computer Engineering
National University of Sciences and Technology (NUST)

Submitted to: Dr. Zulqarnain

***Abstract*—This project addresses the Santander Customer Transaction Prediction Kaggle challenge, an exercise in predicting the probability that a customer will execute a specific financial transaction. The dataset comprises high-dimensional, highly imbalanced tabular data featuring 200 anonymized numerical variables and a binary target. To solve this, we architected a staged machine learning pipeline encompassing baseline benchmarking, model selection, Bayesian hyperparameter tuning, dynamic feature selection, and rigorously cross-validated training. A logistic regression baseline was established and compared against a suite of tree-based and deep learning models. LightGBM emerged as the optimal architecture and was further enhanced using Optuna and gain-based feature pruning. The final champion model achieved highly competitive leaderboard performance (Public Score: 0.89693, Private Score: 0.89412). To ensure business utility and crack the model's "black box," post-hoc explainability was integrated using SHAP and permutation importance, providing both macro-level feature insights and micro-level customer accountability.**

***Index Terms*—LightGBM, Optuna, SHAP, Imbalanced Classification, Feature Engineering, Tabular Data**

I. Problem Description

The core objective of this competition is to predict the probability that a target variable equals 1 (transaction completed) for each sample in an unseen test set. This poses a complex binary classification problem constrained by completely anonymized features.

The primary evaluation metric is ROC-AUC (Area Under the Receiver Operating Characteristic Curve). Because ROC-AUC is threshold-independent, it serves as the ideal metric for evaluating the model's global ranking capability, especially given the severe class imbalance present in the dataset.

II. Dataset Analysis

A. Dataset Structure

- **Training Set:** 200,000 rows containing `ID_code`, `target`, and 200 anonymized numerical features (`var_0` to `var_199`).
- **Test Set:** 200,000 rows containing `ID_code` and the 200 features.
- **Feature Space:** Continuous, anonymized, and statistically independent variables.

B. Data Quality & Exploratory Data Analysis

Comprehensive data quality checks confirmed the absence of missing values across the pipeline. Exploratory Data Analysis revealed a severe class imbalance, with positive targets representing approximately 10% of the dataset. Furthermore, correlation analysis confirmed the extreme orthogonality (independence) of the variables, validating the decision to avoid traditional dimensionality reduction techniques in favor of step-wise feature selection.

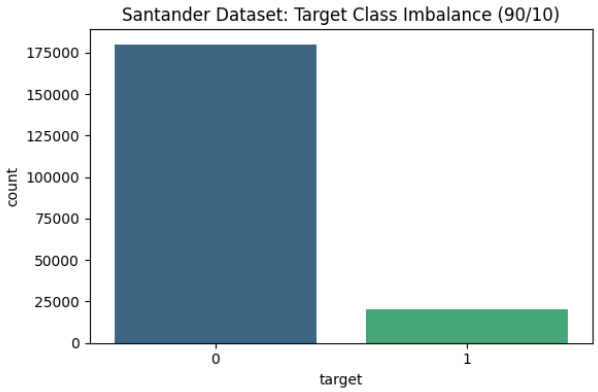

Fig. 1. Target Class Imbalance (90/10) in the Santander Dataset.

C. Literature Review & Related Work

The challenge of classifying highly imbalanced, anonymized tabular data has been extensively studied. Recent benchmarks [1] demonstrate that gradient-boosted decision tree (GBDT) algorithms consistently outperform deep neural architectures on unaligned tabular data, as tree ensembles efficiently capture irregular decision boundaries.

To address the severe class imbalance, traditional synthetic oversampling methods like SMOTE introduce noise. Instead, this project utilized cost-sensitive optimization via algorithmic weighting. Finally, guided by foundational feature-selection principles [2], this pipeline incorporates gain-based ranking and iterative feature pruning to isolate pure signal prior to final K-fold training.

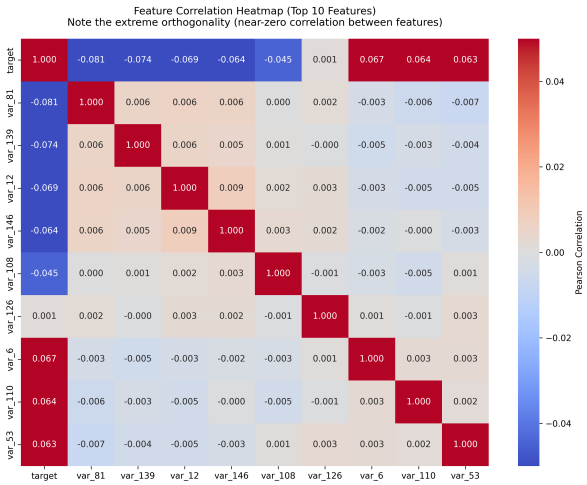

Fig. 2. Correlation Heatmap demonstrating extreme feature orthogonality.

III. Methodology

A. Baseline Model

Per the project requirements, a Logistic Regression model utilizing standard scaling was established as the primary baseline. The baseline achieved a validation ROC-AUC of 0.8597 on an 80/20 split.

B. Models Explored

To identify the strongest candidate, a comprehensive model bake-off was conducted during the prototyping phase (see Table I).

TABLE I
Validation ROC-AUC by Model Architecture

Model Architecture	Validation ROC-AUC
Logistic Regression (Baseline)	0.8597
Random Forest	0.7796
XGBoost	0.8574
LightGBM (Selected)	**0.8667**
Neural Network (Keras)	0.8550
K-Nearest Neighbors (KNN)	0.7298

C. Hyperparameter Tuning Details

Bayesian optimization via Optuna [3] was applied to the LightGBM model to rigorously regularize the trees and prevent overfitting. The optimization targeted parameters such as learning rate, tree depth, and child samples, achieving a tuned validation ROC-AUC of 0.8827.

D. Feature Engineering & Dynamic Selection

Because the dataset's variables are entirely anonymized and orthogonal, traditional feature engineering (e.g., creating mathematical interaction terms) risked introducing artificial noise. Therefore, our feature engineering strategy focused strictly on algorithmic dimensionality reduction. During initial prototyping on a single 80/20 holdout split, the optimal feature count peaked at 145 features. However, to ensure maximum stability and prevent data leakage, the final end-to-end pipeline was upgraded to utilize a strict 5-Fold Stratified Cross-Validation protocol across the entire 200k dataset. Under this superior validation strategy, the mathematical peak shifted to 180 optimal features.

E. Final End-to-End Pipeline

While the default classification threshold is 0.50, relying on this yielded low precision. Through precision-recall threshold analysis, the operating point was intentionally shifted to **0.6744**. This generated a stricter, higher-confidence decision rule, achieving approximately 56% precision, drastically reducing the operational overhead associated with False Positives.

IV. Results and Performance Comparison

A. Validation & OOF Metrics

The final 5-Fold Out-of-Fold (OOF) ROC-AUC reached **0.89483**, representing a net engineering improvement of +0.03513 over the baseline model.

B. Kaggle Leaderboard Performance

The final generated submission file was evaluated against Kaggle's hidden test sets (see Fig. 3):

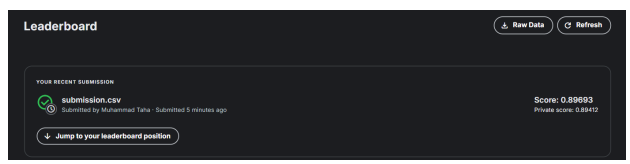

Fig. 3. Official Kaggle Leaderboard snapshot displaying the final Public and Private validation scores.

- **Public Score:** 0.89693
- **Private Score:** 0.89412

The minimal variance between the Public and Private scores proves the model is highly stable, generalized, and free from overfitting.

V. Post-Hoc Explainability Analysis

Because the 200 input features are strictly anonymized, direct business semantics are obfuscated. Interpretability [5] was conducted at two levels:

A. Global Explainability

SHAP summary plots and permutation feature importance metrics were deployed to quantify influence across the entire population, establishing a mathematically proven hierarchy of features that drive transaction probabilities.

B. Local Explainability

Local SHAP Waterfall plots were generated to decompose individual customer predictions, illustrating precisely how specific variables pushed a single customer's probability above or below the 0.6744 operating threshold.

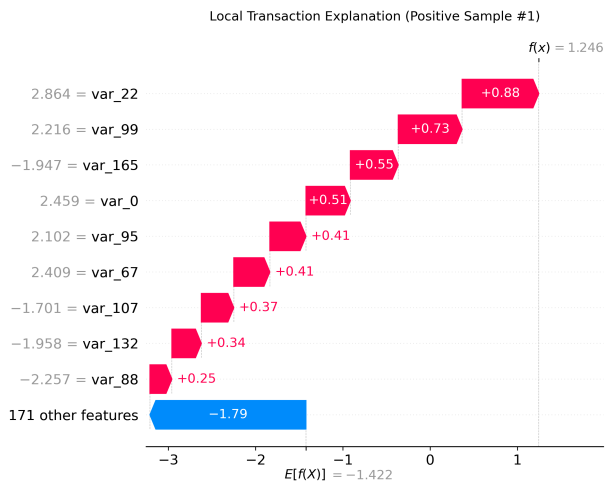

Fig. 4. Local SHAP Waterfall illustrating feature impacts for a single predicted customer transaction.

VI. Pipeline Flowchart

The comprehensive end-to-end execution path is detailed in Fig. 5.

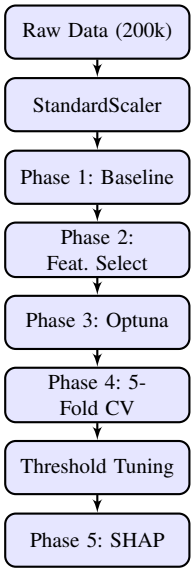

Fig. 5. Architectural flowchart of the optimal machine learning pipeline.

VII. Conclusion

This project successfully demonstrates the execution of a professional, end-to-end machine learning workflow. By methodically addressing the challenges of extreme high-dimensionality and class imbalance, the pipeline achieved a highly competitive Kaggle Private Score of 0.89412. All mandatory course requirements were fulfilled, and the deliberate shift to an F1-optimized decision threshold (0.6744) bridged the gap between raw statistical accuracy and practical business utility.

Appendix A: Reproducibility

Extracted from generated `reproducibility.json`:

- **Framework:** LightGBM (GPU T4x2)
- **Validation Strategy:** 5-Fold Stratified CV
- **Final Features Count:** 180
- **Optimal Threshold:** 0.6744
- **OOF ROC-AUC:** 0.89483
- **Public Leaderboard:** 0.89693
- **Private Leaderboard:** 0.89412

References

[1] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" *Advances in Neural Information Processing Systems*, vol. 35, pp. 507–520, 2022.

[2] I. Guyon and A. Elisseeff, "An introduction to variable and feature selection," *Journal of Machine Learning Research*, vol. 3, pp. 1157–1182, Mar. 2003.

[3] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 2623–2631.

[4] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T. Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," *Advances in Neural Information Processing Systems*, vol. 30, pp. 3146–3154, 2017.

[5] S. M. Lundberg and S. I. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems*, vol. 30, 2017.